

Incident report analysis

Instructions

As you continue through this course, you may use this template to record your findings after completing an activity or to take notes on what you've learned about a specific tool or concept. You can also use this chart as a way to practice applying the NIST framework to different situations you encounter.

Summary	The organization suffered a two-hour denial-of-service attack when a flood of ICMP packets overwhelmed the internal network through an unconfigured firewall. Normal traffic could not reach any resources until the team blocked inbound ICMP, took non-critical services offline, and restored critical ones. Investigation confirmed the vulnerability and led to rate-limiting rules, source-IP verification, monitoring software, and an IDS/IPS. No data was stolen or altered; the impact was temporary loss of availability.
Identify	A volumetric ICMP-flood DoS exploited a missing or overly permissive firewall policy, saturating network resources and blocking all internal connectivity. Affected assets included the entire internal network and the design/marketing systems that depend on it; business processes stopped for two hours. No user credentials were compromised—the gap was purely at the perimeter. Root-cause analysis pointed to the lack of ICMP rate limiting and anti-spoofing controls.
Protect	A new firewall rule now rate-limits inbound ICMP packets and source-IP verification blocks spoofed addresses. The IDS/IPS filters suspicious ICMP characteristics while network-monitoring software watches for abnormal patterns. Firewall policies will be reviewed under change control and staff

	will receive training on reporting performance anomalies. Additional upstream DDoS protection can be considered if volumetric attacks recur.
Detect	Network-monitoring software baselines normal traffic and alerts on sudden ICMP volume spikes or single-source floods. The IDS/IPS generates real-time alerts on suspicious ICMP traits and feeds them into centralized logging. Firewall logs are checked for rate-limit hits and verification failures, with thresholds set so staff are notified within minutes. Periodic tests of the detection rules confirm they remain effective.
Respond	Future incidents will be contained by immediately tightening or blocking inbound ICMP at the firewall and isolating non-critical services if needed. The IDS/IPS and source verification neutralize ongoing floods while logs are collected to identify sources, volume, and any spoofing. Leadership and affected teams are notified of status; law enforcement is engaged only if required. After resolution the playbook is updated and a tabletop exercise is scheduled.
Recover	Future incidents will be contained by immediately tightening or blocking inbound ICMP at the firewall and isolating non-critical services if needed. The IDS/IPS and source verification neutralize ongoing floods while logs are collected to identify sources, volume, and any spoofing. Leadership and affected teams are notified of status; law enforcement is engaged only if required. After resolution the playbook is updated and a tabletop exercise is scheduled.

Reflections/Notes: Basic perimeter hardening and continuous monitoring would have shortened or prevented the outage. Applying the NIST CSF functions systematically strengthens both prevention and response for similar availability threats.